Mississippi Valley State University

CS 341 — Discrete Structures

Lecture 29: Settling a Question From Week 1

Other Starting Points, Inequalities, and Divisibility

Chartrand & Zhang, Section 4.2, pp. 123–128

Name: _____ Date: _____

Unfinished business

Practice Set 1, Problem 15 asked about $2^n > n^2$. You found it true at $n = 1$, false at $n = 2, 3, 4$, and true for every value you tried from 5 on — and you *could not prove it*, because checking cases is not proof.
Today you prove it. It takes about six lines.

By the end of today you should be able to

1. Run an induction proof from a starting point other than 1.
2. Prove an inequality by induction.
3. Prove a divisibility statement by induction.

1. Starting somewhere else

The Principle, with base m

For a fixed integer m, let $S = \{i \in \mathbb{Z} : i \geq m\}$. Then $\forall n \in S,\ P(n)$ is true if
(1) $P(m)$ is true, and
(2) $\forall k \in S,\ P(k) \Rightarrow P(k+1)$ is true.

Same method; the chain simply starts at m instead of 1. When $m = 1$ this is the original principle.

2. An inequality: $n! > 2^n$

n	1	2	3	4	5	6
$n!$	1	2	6	24	120	720
2^n	2	4	8	16	32	64

False for $n \leq 3$, true from $n = 4$. So the base step is at 4.

Result

For every integer $n \geq 4$, $\quad n! > 2^n$.

Proof. We proceed by induction. **Base step:** $4! = 24 > 16 = 2^4$.

Inductive step. Assume $k! > 2^k$ for an integer $k \geq 4$. Then

$$(k+1)! = (k+1)\,k! \;>\; (k+1)\,2^k \;>\; 2 \cdot 2^k = 2^{k+1}.$$

The first inequality uses the inductive hypothesis; the second uses $k + 1 > 2$, true since $k \geq 4$.

By induction, $n! > 2^n$ for every $n \geq 4$. ■

How inequality proofs run

Start with one side of $P(k+1)$, use the inductive hypothesis to bring in the k version, then chain inequalities to the other side. **Each link must be justified**, and one of them is always the hypothesis.

3. The Week 1 question, answered

Result

For every integer $n \geq 5$, $\quad 2^n > n^2$.

Proof. We proceed by induction. **Base step:** $2^5 = 32 > 25 = 5^2$.

Inductive step. Assume $2^k > k^2$ for an integer $k \geq 5$. Then

$$2^{k+1} = 2 \cdot 2^k \;>\; 2k^2.$$

It remains to show $2k^2 \geq (k+1)^2 = k^2 + 2k + 1$, that is, $k^2 \geq 2k + 1$.

Since $k \geq 5$, we have $k^2 = k \cdot k \geq 5k = 2k + 3k \geq 2k + 15 > 2k + 1$.

So $2^{k+1} > 2k^2 \geq (k+1)^2$. By induction, the result holds for every $n \geq 5$. ■

Why the base step had to be 5

The inductive step only needed $k \geq 3$ to go through. But the claim is *false* at $n = 2, 3, 4$, so no chain could start before 5.

The base step decides where the result begins. Choosing it is part of the proof, and it is set by where the claim is actually true — which is why testing small values first is not optional.

4. Divisibility

Claim

For every positive integer n, $\quad 3$ divides $4^n - 1$.

Proof. We proceed by induction. **Base step:** $4^1 - 1 = 3$, which is divisible by 3.

Inductive step. Assume $3 \mid 4^k - 1$, so $4^k - 1 = 3m$ for some integer m. Then

$$4^{k+1} - 1 = 4 \cdot 4^k - 1 = 4(4^k - 1) + 3 = 4(3m) + 3 = 3\big(4m + 1\big).$$

Since $4m+1$ is an integer, $3 \mid 4^{k+1}-1$. By induction the claim holds. ■

The trick

The goal was to make 4^k-1 appear so the hypothesis could be used. Writing $4\cdot 4^k-1 = 4(4^k-1)+3$ is **adding and subtracting the right thing** — the same instinct as splitting off the last term of a sum. Find where the k case lives inside the $k+1$ case.

Exit ticket

1. For the claim $3^n \geq 2n+1$ for all $n \geq 1$, what is the base step?

2. In the proof that $2^n > n^2$, why is the base step 5 and not 1?

Before Friday

Read Section 4.3, "Sequences," pp. 128–137, and Section 4.4, "The Strong Principle," pp. 137–142.

Self-check (not collected). Section 4.2, Exercises 1, 3, and 5 — odd-numbered, answers in the back.

Homework 10 is due Friday, October 30, 11:59 p.m.

Questions I still have